

Project Report

Geometry-Aware Computational Framework for Prioritizing Residues Involved in Protein–Protein Interfaces

Project Scope and Deliverables

This project develops and evaluates a **structure-based, geometry-aware computational framework** for protein–protein interaction (PPI) interfaces. Rather than attempting to fully annotate all interface residues, the goal is to **rank residues on a protein chain by their likelihood of participating in a specific protein–protein interface**, using only local 3D structural information.

The emphasis is on producing **interpretable, structure-level outputs**—both quantitative rankings and 3D visualizations—that allow to reason about *where* and *why* a protein surface appears interface-like.

Concrete deliverables

1. Curated structural dataset

- Approximately 1,900 protein-chain examples derived from experimentally resolved PDB protein–protein complexes
- Residue-level interface labels generated using a transparent geometric proximity definition
- Train/validation/test splits performed at the **complex level** to prevent structural leakage

2. End-to-end preprocessing pipeline

- mmCIF parsing and chain-level extraction
- Construction of residue-level spatial graphs
- Interface labeling at multiple distance thresholds (4 Å, 5 Å, 6 Å)
- Dataset statistics and interface-size distributions used for quality control and analysis

3. Predictive models of increasing complexity

- A distance-only geometric heuristic (sanity-check baseline)
- Classical machine-learning models (logistic regression, random forest) using residue-level features
- A compact geometry-aware graph neural network (GNN) operating on residue graphs

4. Quantitative benchmarking of residue prioritization

- PR-AUC to assess global ranking quality under strong class imbalance
- Precision@K and F1@K (K = 10, 20, 30) to reflect realistic residue inspection budgets
- Direct comparison across heuristic, classical ML, and GNN-based approaches

5. 3D structural case studies

- Visualization of full ground-truth interface regions for context
- Visualization of Top-K model predictions mapped onto protein structures
- Spatial analysis of true positives, false positives, and false negatives under a fixed prioritization budget

6. Interpretation-focused analysis

- Examination of when local geometry is sufficient to identify interface-like regions
- Identification of common failure modes visible in structural space

- Discussion of biological and modeling limitations inherent to geometry-only inference
-

1. Biological and Computational Motivation

Protein–protein interfaces typically consist of **spatially localized surface patches** formed by multiple residues with complementary geometry and chemistry. In practice, many experimental and computational workflows do not require an exhaustive enumeration of all interface residues. Instead, the more common question is:

“Which residues or surface regions should be examined first?”

This is especially relevant for:

- exploratory structural analysis
- hypothesis generation
- guiding mutagenesis or further modeling

Accordingly, this project frames interface prediction as a **residue prioritization problem**, rather than a strict binary classification task.

From a computational perspective, this framing avoids brittle probability thresholds and aligns more naturally with:

- strong class imbalance (few interface residues relative to total residues)
- large variability in interface size across complexes
- uncertainty introduced by distance-based geometric labeling

By combining ranking-based evaluation with explicit 3D visualization, the project aims to connect **quantitative performance** with **structural intuition**, which is often critical for biological interpretation.

2. Interface Label Definition

Interface residues are defined using a standard geometric proximity rule:

A residue is labeled as interfacial if **any heavy atom** of that residue lies within **5 Å** of **any heavy atom** of the partner chain.

This definition:

- is widely used in structural bioinformatics
- is directly computable from atomic coordinates
- captures physical proximity rather than biochemical importance

The project explicitly treats these labels as **approximate geometric indicators**, not as definitive functional annotations.

3. Dataset Construction and Characteristics

Data source and filtering

Protein structures were obtained from the Protein Data Bank (PDB) in **mmCIF format**, which supports detailed atomic and chain-level metadata.

Processing steps included:

- restriction to protein–protein complexes
- removal of ligands, nucleic acids, and solvent
- exclusion of hydrogen atoms
- filtering of non-standard residues
- removal of chain pairs with trivial interfaces

Each example corresponds to a **single protein chain** drawn from a known interacting pair. The partner chain is used **only to generate interface labels**, not as part of the model input.

Dataset scale and variability

After filtering:

- ~2,100 valid chain-pair examples were identified
- ~1,900 were retained after interface-size filtering

Interface sizes vary widely, ranging from a handful of residues to large multi-patch interfaces, introducing substantial heterogeneity and class imbalance.

4. Structural Representation and Features

Residue-level graph representation

Each protein chain is represented as a **residue-level spatial graph**:

- Nodes correspond to amino-acid residues
- Edges connect residues within a fixed intra-chain distance cutoff
- Edge features encode Euclidean distance between residues

This representation captures **local 3D neighborhood structure** while remaining computationally lightweight and interpretable.

Node features (structure-only)

Each residue is described using features derived solely from the 3D structure:

- Amino-acid identity (one-hot)
- Coarse physicochemical categories (e.g., hydrophobic, charged, aromatic)
- Solvent exposure proxy
- Local contact density
- Relative sequence position within the chain

No evolutionary profiles, sequence alignments, or pretrained embeddings are used, ensuring that all information comes from **single-structure geometry**.

5. Models Evaluated

Surface-exposure heuristic (Baseline 0)

A purely geometry-based baseline is used to establish a lower bound on performance without learning or partner-specific information. This heuristic operates on a **single protein chain** and ranks residues based on their **local packing density** within the chain.

Residues are represented as nodes in an intra-chain residue graph, with edges connecting spatially neighboring residues. For each residue, a local contact density is computed as the number of neighboring residues within a fixed distance threshold. Only residues in the lowest quantile of contact density (corresponding to surface-exposed residues) are retained, and these residues are ranked inversely by their contact density.

This baseline captures the intuition that interface residues must be **accessible and exposed**, while ignoring all information about the binding partner. It therefore represents a **true partner-agnostic, geometry-only reference point**. Performance relative to this baseline indicates whether learned models extract meaningful information beyond simple surface exposure.

Classical machine-learning models

Logistic regression and random forest models were trained using the same residue-level features.

Observed behavior:

- logistic regression struggled to model nonlinear spatial interactions
- random forest models often became overly conservative, achieving high precision at the cost of very low recall

Geometry-aware graph neural network

The main model is a **small, residue-level GNN** with:

- three message-passing layers
- distance-weighted neighbor aggregation
- binary cross-entropy training objective

The model outputs a **continuous interface propensity score** for each residue, reflecting its likelihood of being near the interaction interface given local structural context.

The architecture is intentionally simple and does not include attention mechanisms, cross-chain message passing, or pretrained components.

6. Evaluation Strategy and Top-30 Prioritization

Training vs. evaluation

While the model is trained as a per-residue binary classifier, evaluation is performed in a **ranking-based manner**. This avoids unstable probability thresholds and aligns with practical usage.

Residues are ranked by predicted interface probability.

Why Top-30?

Top-30 prioritization is used as a **fixed, biologically reasonable inspection budget**:

- Many protein-protein interfaces involve tens of residues per chain
- Experimental and structural follow-up typically focuses on a limited subset
- Top-K metrics allow comparison across proteins of different lengths

Top-30 does not imply that the interface consists of exactly 30 residues; it is a controlled evaluation lens for measuring enrichment.

7. Quantitative Results

Test-set comparison (representative values)

- **Surface-exposure heuristic**
 - PR-AUC ≈ 0.35
 - Precision@30 ≈ 0.38
 - F1@30 ≈ 0.23
- **Logistic regression**
 - PR-AUC ≈ 0.33
 - High variance in Precision@K
 - Very low F1 due to unstable thresholding
- **Random forest**

- PR-AUC ≈ 0.35
- Precision@30 ≈ 0.50
- F1@30 ≈ 0.01 (severely conservative)
- **Geometry-aware GNN**
 - PR-AUC ≈ 0.36 – 0.38
 - Precision@30 ≈ 0.44 – 0.47
 - F1@30 ≈ 0.26 – 0.32

The GNN provides **modest but consistent enrichment** of interface residues relative to baselines.

8. Structural Case Studies (3D Visualizations)

The 3D visualizations are a central component of this project, as they reveal *how* the model's residue rankings manifest in real protein structures beyond what aggregate metrics can show.

Each case study presents two complementary views:

1. **Ground-truth interface view**, showing all residues labeled as interfacial based on the geometric distance criterion.
2. **Top-30 prediction view**, showing only the 30 residues with the highest predicted interface propensity, color-coded by correctness (true positives, false positives, false negatives).

Across examined cases, several consistent spatial patterns emerge.

First, **true positive predictions tend to cluster into spatially coherent surface patches** rather than appearing as isolated residues. In successful cases, the top-ranked residues form contiguous regions on the protein surface that overlap with known interface areas. This indicates that the model is not simply detecting individual residues in isolation, but is responding to **local geometric context** characteristic of interaction surfaces.

Second, **false positives are typically located on exposed surface regions with interface-like geometry**. These residues often lie on protruding loops or flat surface patches that are solvent-exposed and locally dense, but do not participate

in the specific interaction present in the crystal structure. From a structural biology perspective, these false positives are plausible candidates: they are not buried core residues, but rather surface residues that could conceivably participate in interactions under different contexts. This behavior is consistent with a geometry-only model that lacks partner-specific information.

Third, **false negatives are dominated by interface residues that fall outside the Top-30 cutoff**, particularly in complexes with large interfaces. In such cases, the ground-truth interface may involve 50–100 residues, making it impossible for a fixed Top-30 prioritization to recover all interface residues by design. Importantly, these false negatives are often spatially adjacent to true positives, indicating that the model partially captures the interface region but must truncate its predictions due to the ranking constraint.

Overall, the visualizations show that the model tends to identify **core or prominent interface regions**, while deprioritizing peripheral or less distinctive interface residues. This behavior aligns with the intended use of the model as a prioritization tool rather than a full interface annotator.

9. Interpretation and Limitations

The model should be interpreted strictly as a **geometry-based residue prioritization method**, and its outputs should be read accordingly.

From a practical standpoint, the model answers the question:

Which residues on this protein chain appear most interface-like, given their local 3D environment?

It does **not** attempt to determine:

- whether an interface exists in the absence of a known partner,
- which partner a protein may bind,
- or which residues are energetically or functionally critical for binding.

The visualizations make clear that the model's predictions are driven by **local surface geometry and neighborhood structure**. This explains both its strengths and its limitations. Geometry alone is often sufficient to highlight plausible interaction regions, but it cannot disambiguate between multiple exposed surface

patches that share similar geometric characteristics. As a result, some false positives are structurally reasonable but biologically irrelevant for the specific complex.

Additionally, because the model operates at the single-chain level and does not incorporate partner-chain features during inference, it cannot account for **partner-specific complementarity**. This limitation is visible in cases where predicted residues lie close to, but not directly within, the true interface region.

These constraints are not implementation artifacts, but rather intrinsic to the chosen problem formulation and feature set.

10. Conclusion

This project demonstrates that local 3D structural context, modeled using a compact and interpretable graph neural network, can be used to **prioritize protein residues that are enriched for participation in protein–protein interfaces**.

Quantitatively, the model provides modest but consistent improvements over distance-only and classical machine-learning baselines in ranking-based metrics. Qualitatively, the 3D visualizations show that the model tends to highlight **coherent surface regions** that overlap with known interfaces, while producing biologically plausible false positives on exposed surfaces.

The primary value of the project lies not in achieving high predictive accuracy, but in establishing a **clear, reproducible, and biologically interpretable framework** for structure-based residue prioritization. The visual outputs, in particular, make the model's behavior transparent and facilitate direct structural reasoning by biologists.

As such, this work serves as a solid foundation for future extensions that incorporate partner-specific geometry, evolutionary conservation, or more expressive geometric representations, while remaining grounded in realistic structural biology use cases.

Visualized Output

Visualization definitions:

Views:

- View 1 (all red) visualizes the **ground truth**. It shows all interface residues
- View 2 visualizes the model **predictions**. It consists of **green colored** residues (predicted interface residues that actually ARE interface residues or TPs), **orange colored** residues (predicted interface residues that actually ARE NOT interface residues or FPs), **purple colored** residues (not predicted interface residues that actually ARE interface residues, FNs)

Interpreting the Visualized Outputs:

The visualizations compare **ground-truth interface residues** with **Top-30 model predictions**, and should be interpreted as **residue prioritization**, not full interface prediction.

- **Ground truth view:** shows *all* residues labeled as interface based on the geometric distance criterion ($\leq 5 \text{ \AA}$). Interface size therefore varies across proteins and may exceed 30 residues.

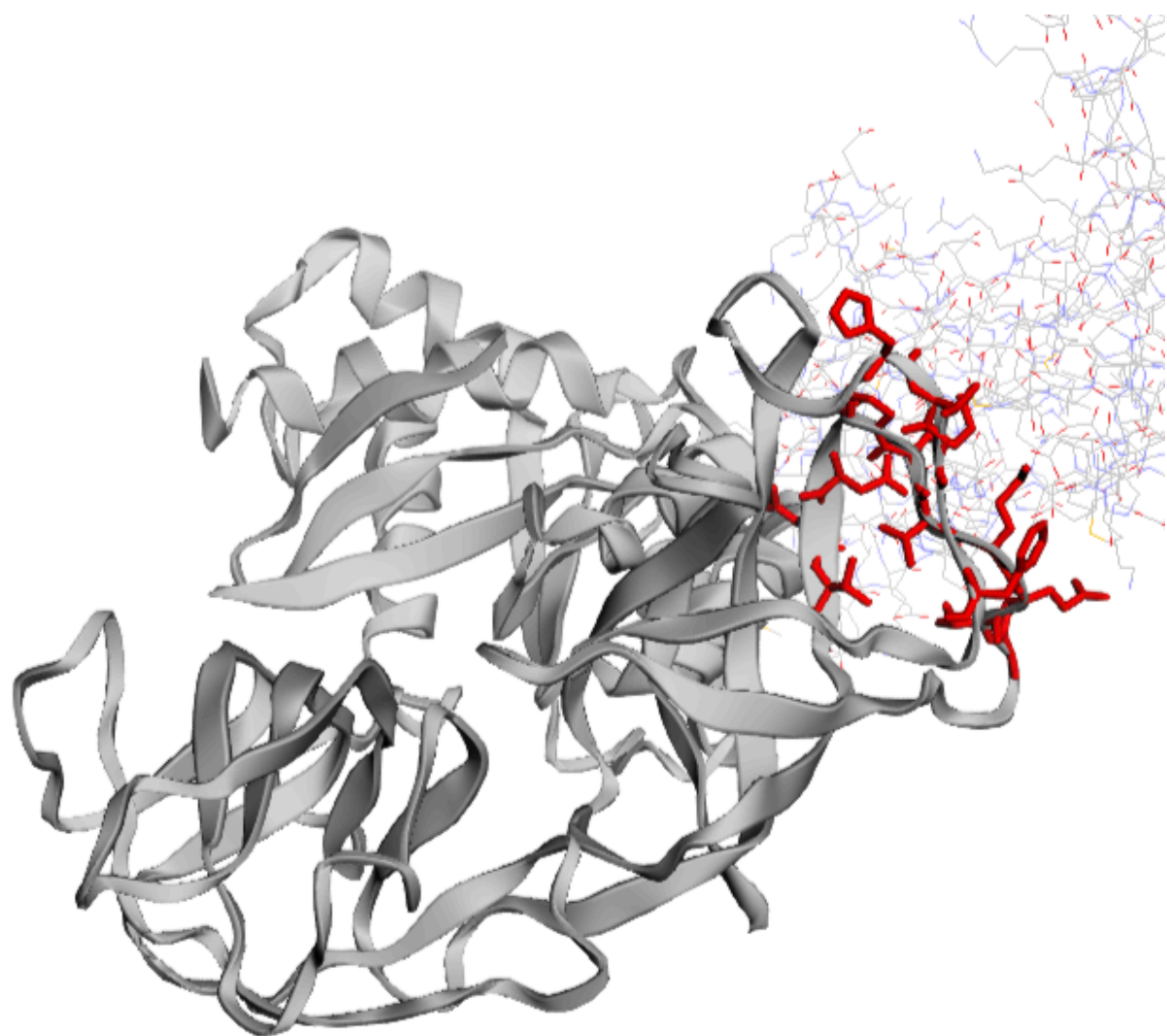
- **Prediction view:** shows **only the Top-30 residues** ranked highest by the model's predicted interface probability. The model assigns scores to *all* residues, but only the top-ranked subset is visualized.

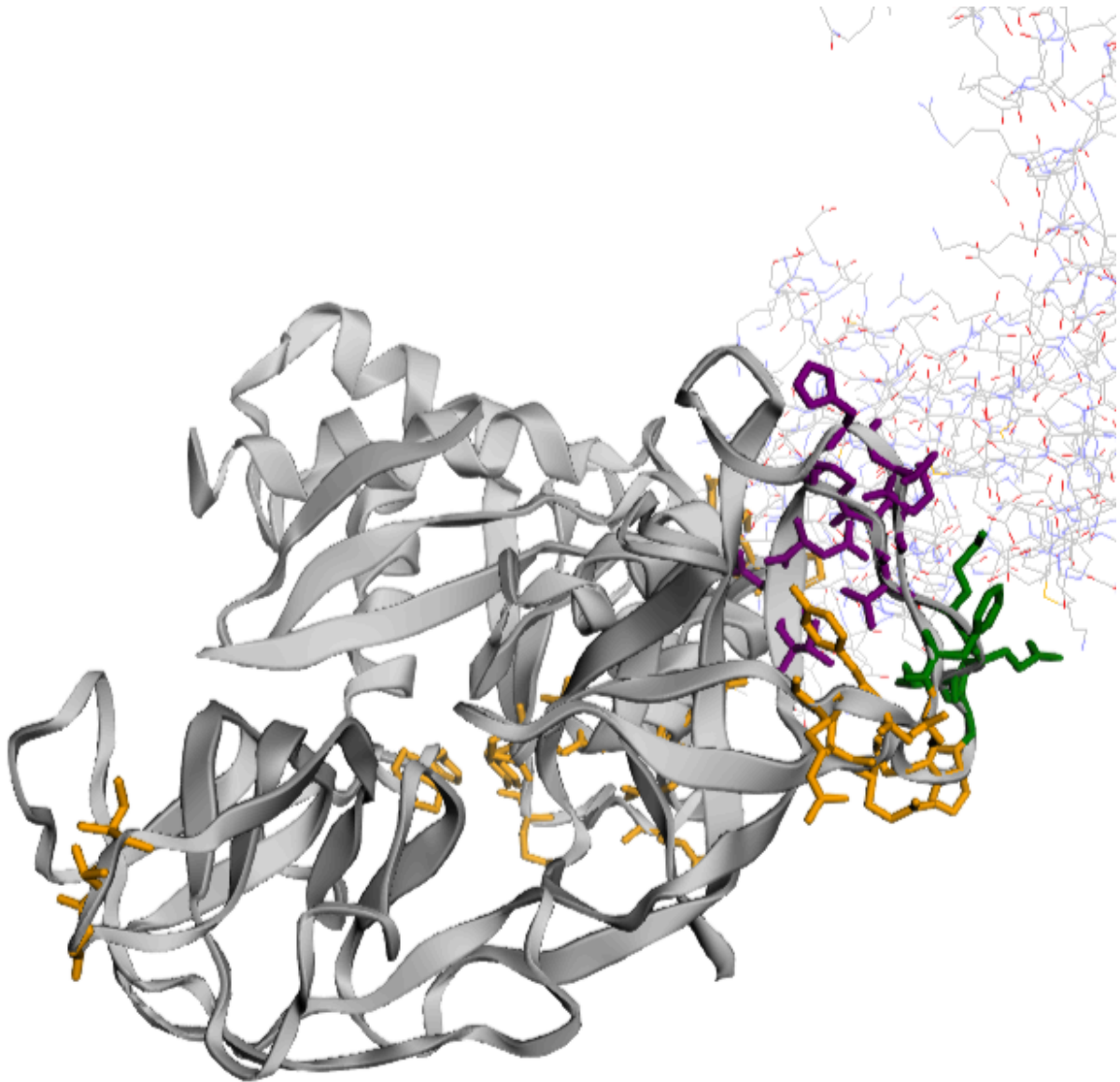
Because the model is designed to **prioritize a fixed number of candidate residues**, it is expected that many ground-truth interface residues are not shown among predictions, especially for large interfaces. These are counted as false negatives relative to the Top-30 cutoff, not as misclassified residues.

Top-30 prioritization reflects a realistic inspection budget and enables fair comparison across proteins of different sizes. The purpose of the visualization is to assess whether the top-ranked residues cluster in plausible interface regions, rather than whether the entire interface is recovered.

Test Example 1

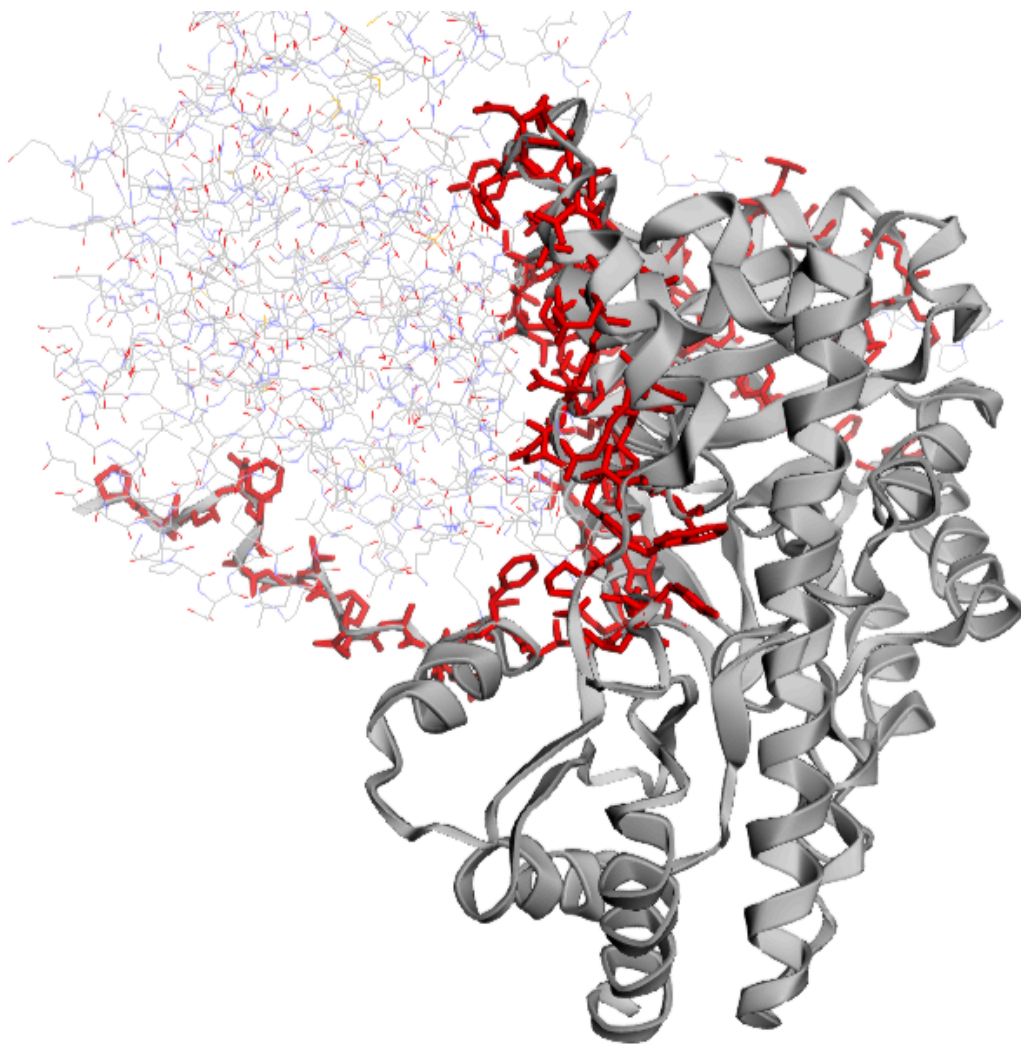
CASE 8: 1AIP | chain F (side=A) | N=372 | GT_pos=14
TopK=30: TP=5 FP=25 FN=9 (FN is vs ALL GT positives)

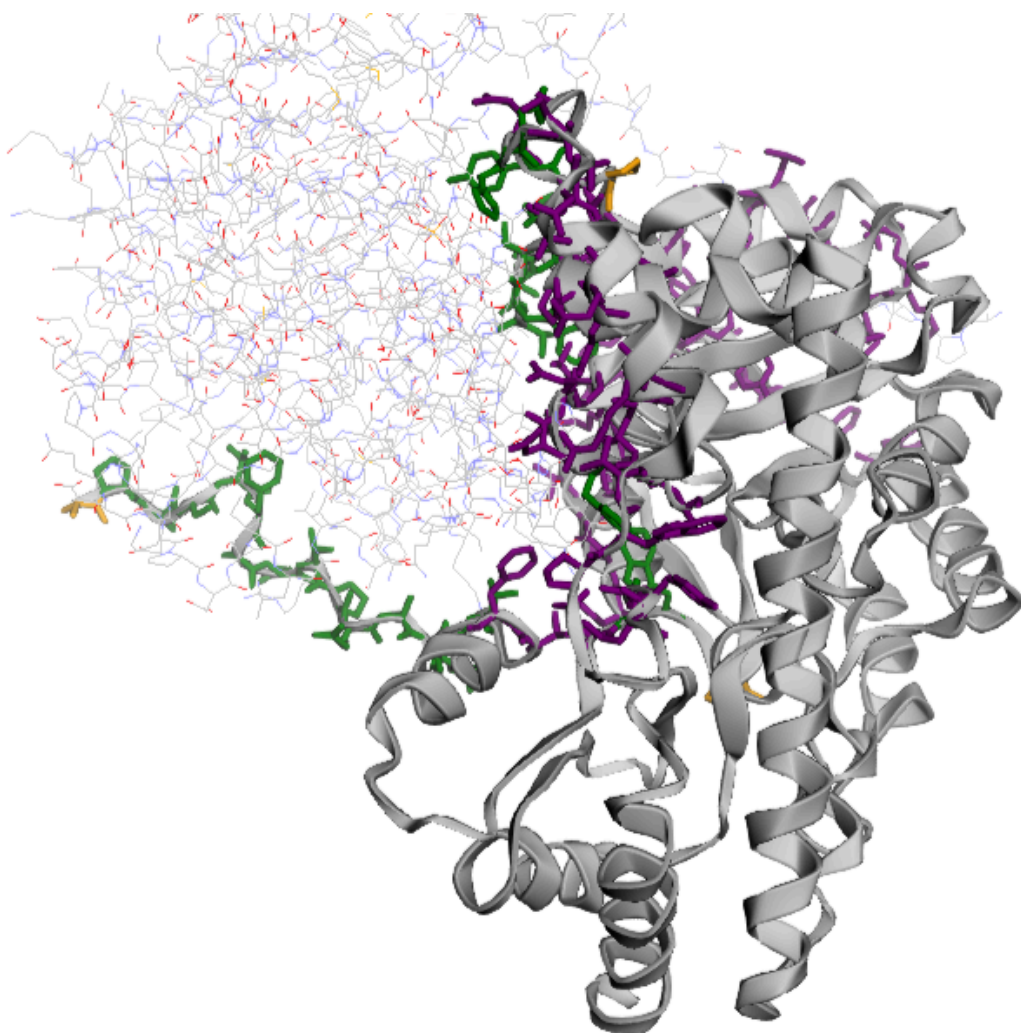




Test Example 2

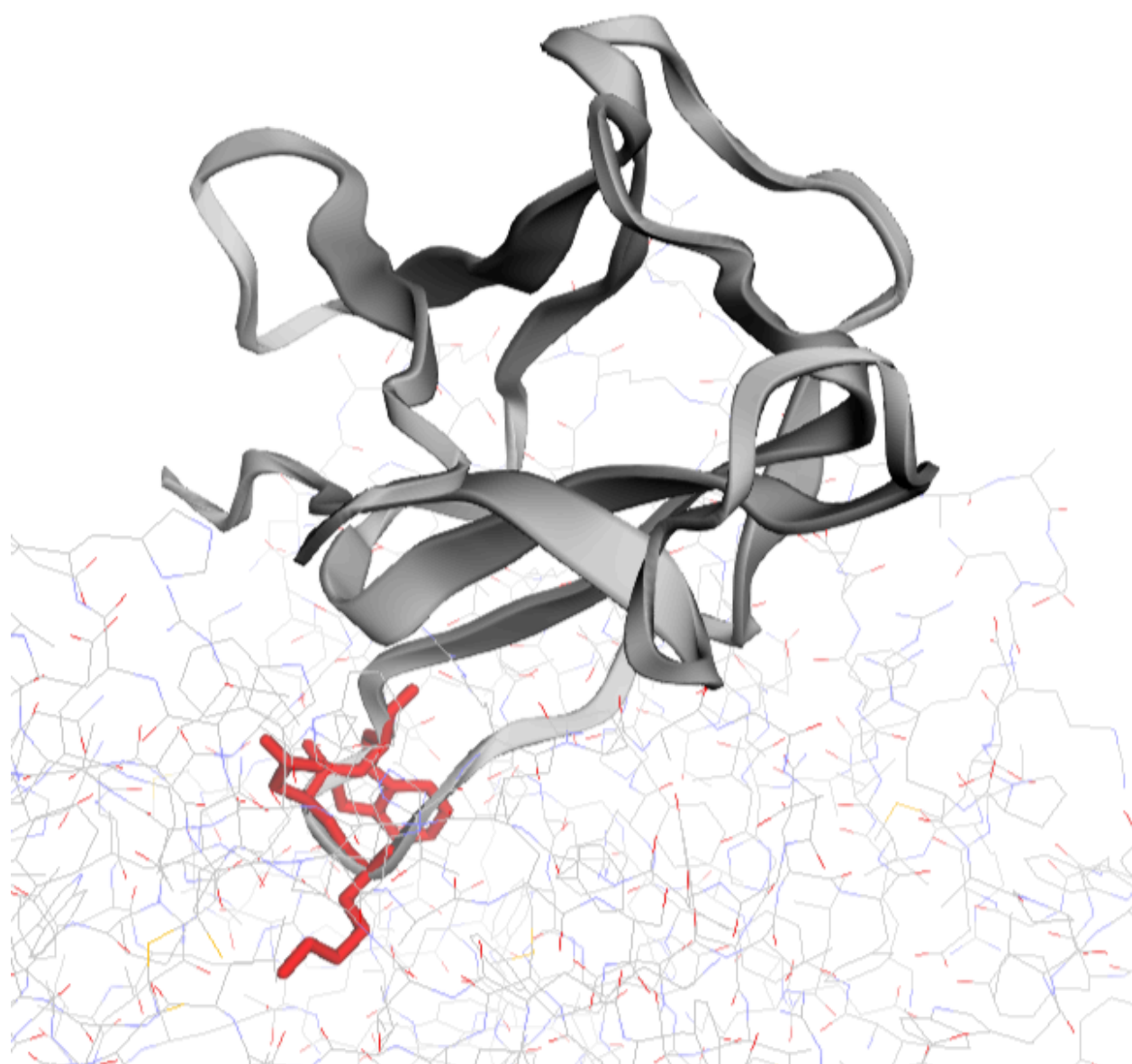
CASE 17: 1AJS | chain A (side=A) | N=412 | GT_pos=89
TopK=30: TP=26 FP=4 FN=63 (FN is vs ALL GT positives)

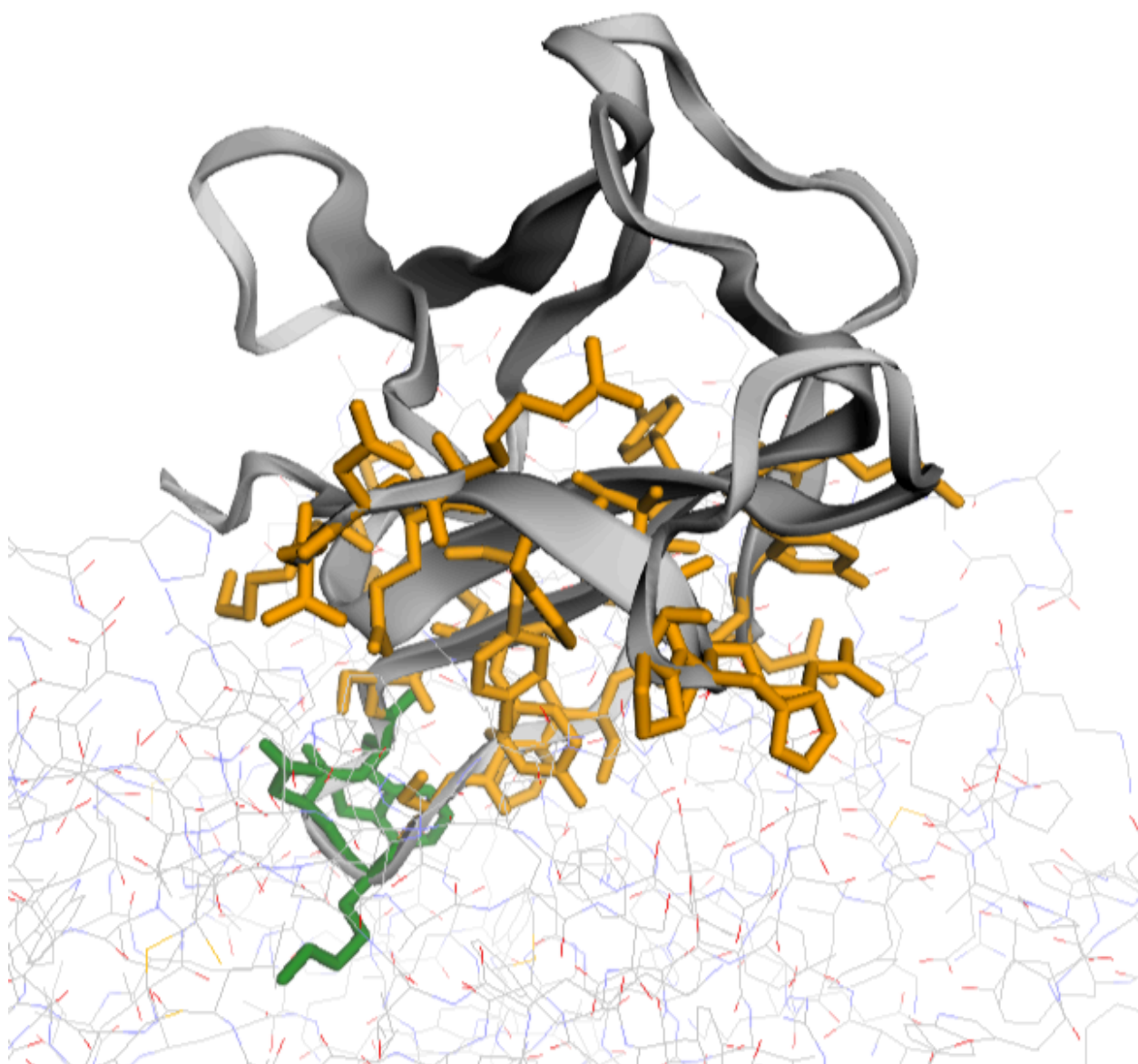




Test Example 3

CASE 25: 1AKJ | chain B (side=A) | N=98 | GT_pos=3
TopK=30: TP=3 FP=27 FN=0 (FN is vs ALL GT positives)





Test Example 4

CASE 20: 1AK4 | chain C (side=A) | N=145 | GT_pos=10
TopK=30: TP=6 FP=24 FN=4 (FN is vs ALL GT positives)

